UNIVERSITY OF THE
WITWATERSRAND,
JOHANNESBURG

Human Research Ethics Committee: (Medical)
FWA Registered No IRB 00001223

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EMAILED

28 June 2022

Prof MF Chersich,

Wits RHI Head Quarters
Hillbrow Health Precinct
22 Esselen Street
Hillbrow
2001

Email: mchersich@wrhi.ac.za

CC: Mr CJ Parker
Wits RHI - Esselen Street
Hugh Solomon Building
Hillbrow Health Precinct, 22 Esselen Stre
Hillbrow
2001

Email: cparker@wrhi.ac.za

Dear Prof Chersich,

PROTOCOL: HEAT002 : INNOVATIVE MACHINE LEARNING AND MULTI-SOURCE DATA ANALYSIS TOWARDS DEVELOPMENT OF AN URBAN HEAT-HEALTH EARLY WARNING SYSTEM FOR AFRICAN CITIES

RE: ASSESSORS QUERIES

REFERENCE NO: 220606

Please would you be so kind as to respond to the following queries raised regarding the above Protocol:

1. AT THE RECENT HUMAN RESEARCH ETHICS COMMITTEE: (MEDICAL) (HREC) meeting held on 24 June 2022, the committee agreed that the above-mentioned study can be approved. There were no points that required clarification.

Your response to the above queries raised would be appreciated on or before 18 July 2022.

Yours sincerely

MRS KIM ZARA MOTHIBA

For and on behalf of the Human Research Ethics Committee: (Medical)

1. 15 JUNE 2022 - HEREWITH SUMMARY OF REVIEW AND QUERIES RAISED BY ETHICS ASSESSOR, PROF C SMITH - TEL: 011 488 3487

SUMMARY OF REVIEW
The overarching goal of this Research Study is to advance understanding of complex spatially and demographically stratified heat-health interactions in large African cities and to apply this information to develop locally relevant and risk-stratified Early Warning Systems.

More specifically, this overarching goal will be pursued through three aims:

1)Map intra-urban heat vulnerability and exposure across urban areas in large African cities (Aim1)
2) a) Quantify intra-urban socio-economic and environmental vulnerability (Aim 1a)
3) Develop high resolution urban temperature hazard maps (Aim 1b)
4) Develop a spatially and demographically stratified heat-health outcome forecast model in order to predict the probability of adverse health outcomes at different temperature thresholds (Aim 2)
5) Develop an Early Warning System reflective of geospatial and individualized risk patterns (Aim 3)
6) Develop an EWS based on a digital App (Aim 3a)
7) Develop a "tiered" EWS in collaboration with the Ministries of Health, National Meteorological and Hydrological Services, and other relevant stakeholders (Aim 3b).

Study site JHB South Africa and Abidjan Ivory Coast.

The project raises several ethical and legal considerations. This includes:
1) the use of secondary data for research purposes that may differ from the original purpose of the study,
2) sharing data across countries in Africa, and
3) the presence of potential identifying information.
No new data will be collected, all data used will be secondary.

The primary health datasets will be a collation of datasets containing longitudinal data generated in studies that began enrolling participants in 2010 or thereafter (in Johannesburg and in Abidjan).

The datasets are drawn from trials among HIV-infected adults, HIV-uninfected adults and adults participating in trials related to COVID-19 prevention or treatment.

These datasets will become available throughout the project and will be added based on amendments to original agreements on each of these trials and ethics approval.

We will include all participants and associated data in the trial and not use sampling to select the data.
The following databases will be included:

a) databases that contain population groups of interest (High quality health data which can be mapped onto Climate data)
b) the investigators of the original studies have given permission for use of the data.

There is a minimal risk of breech of confidentiality. Risk to a participant in this study are minimal as there are no new procedures involved and many steps will be taken to protect their personal information.

Data from participants will be safeguarded and only made available to study investigators through strict access control platforms.
Personal identifying information such as date of birth or address will further, only be made available to a smaller subset of investigators who will be responsible for anonymising the data through various methods.

Moreover, no personal details of any participant will be made publicly available through this project or published in a journal article.
Data may not be fully unidentifiable as we are collecting date of birth and location data, where available. However, we will take steps to minimise the risk to privacy of participants.

We will not be collecting names of participants and no identifiable data will be published.

In addition, the data will be safeguarded in a password protected server with limited access. In addition, we will use principles of minimisation, to ensure that only essential data that is required for the study objectives are stored. Lastly, wherever possible, data will be anonymised.

There are no points that require clarification

IN VIEW OF THE ABOVE, PROF SMITH RECOMMENDS THAT THE PROTOCOL BE APPROVED

Your response to the above queries raised would be appreciated on or before 18 July 2022.

Yours sincerely

MRS KIM ZARA MOTHIBA

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22 June 2022

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Dear Prof Chersich,

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RE: ASSESSORS QUERIES

REFERENCE NO: 220606

Please would you be so kind as to respond to the following queries raised regarding the above Protocol:

1. 21 JUNE 2022 - HEREWITH SUMMARY OF REVIEW AND QUERIES RAISED BY ETHICS ASSESSOR, PROF RGM THOM - TEL: 011 350 0079

   SUMMARY OF REVIEW: The aim of this project is to advance understanding of complex spatially and demographically stratified heat-health interactions in large African cities in order to develop early warning systems for negative heat-health interactions. This is a secondary analysis of existing databases of individual participant data in large studies which were conducted between January 2000 and June 2022. Two cities will be studies, Abidjan and Johannesburg. This application relates to the Johannesburg. Studies will include HIV related and COVID-related studies, and will include a range of participants (HIV-positive and negative and COVID positive and negative). Various health-related variables will be extracted from these studies and mapped against detailed environmental and socio-demographic information, to determine associations which might help to develop early warning systems for negative heat impacts in the future.

   This is a well-prepared application with attention to ethical issues. Various measures are in place to protect against breach of confidentiality and to aggregate information so as to de-identify personally identifying information that may be accessed during the course of project.

   There are no pints that require clarification

   IN VIEW OF THE ABOVE, PROF THOM RECOMMENDS THAT THE PROTOCOL BE APPROVED

Your response to the above queries raised would be appreciated on or before 18 July 2022.

Yours sincerely

MRS KIM ZARA MOTHIBA

For and on behalf of the Human Research Ethics Committee: (Medical)